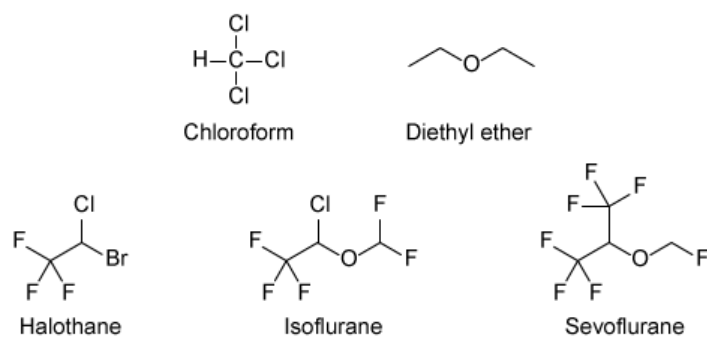


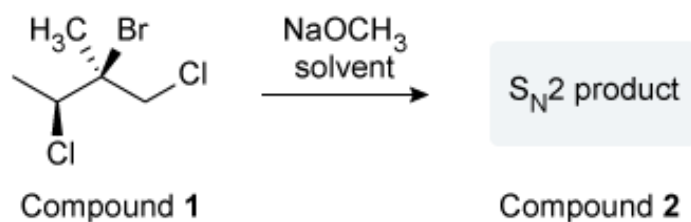
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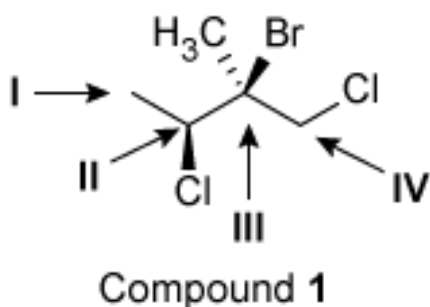
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Scheme I

Which carbon atom in Compound 1 will serve as the major electrophile for the substitution reaction in Scheme I?



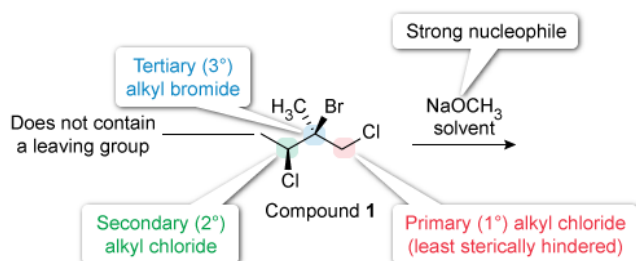
- ☐ A. I [3%]
- ☐ B. II [9%]
- ☐ C. III [22%]
- ✓ ☒ D. IV [64%]

Correct

64% answered correctly

Explanation:

Selecting the best electrophile for an $\text{S}_{\text{N}}2$ reaction



$\text{S}_{\text{N}}2$ reactions occur the fastest for less sterically hindered carbon electrophiles

In a substitution reaction, a functional group on an atom is displaced by a different functional group. The lost functional group is the **leaving group**, named for its ability to leave with the pair of electrons from the sigma bond that connected it to the rest of the molecule. Good leaving groups are typically weak bases that can distribute the additional electron density through either a large surface area or resonance-delocalization. The leaving group polarizes this sigma bond to generate an electron-deficient carbon atom (an **electrophile**). The displacing group (a **nucleophile**) contains a lone pair of electrons that it donates to the electrophilic carbon atom to form the new sigma bond.

An **S_N2 reaction** is a second-order substitution that takes place in a single, concerted reaction step between the electrophile and nucleophile. S_N2 reactions tend to require a **strong nucleophile** and occur the fastest with **sterically unhindered electrophiles**; tertiary electrophiles are unable to participate in S_N2 reactions.

In this question, the oxygen atom in sodium methoxide (NaOCH₃) is a strong nucleophile and will attack the most sterically accessible carbon atom that contains a good leaving group (chlorine or bromine). Given that position IV is a primary alkyl halide and is bonded to only one other carbon atom, **position IV** is the least sterically hindered electrophilic carbon atom for an S_N2 reaction with NaOCH₃.

(Choice A) A substitution reaction requires the electrophilic carbon atom to be bonded to a leaving group. Because carbon I is not bonded to a halogen atom, it does not have a leaving group and will not be able to react with NaOCH₃.

(Choice B) S_N2 reactions occur at a faster rate with less-substituted electrophiles. Carbon II is a *secondary* alkyl halide and is *not* the least-substituted electrophile in Compound 1.

(Choice C) While position III is bonded to the best potential leaving

group (bromine), it is a *tertiary* alkyl halide unable to react with nucleophiles through an S_N2 reaction.

Educational objective:

S_N2 reactions occur between strong nucleophiles and less-substituted electrophilic carbon atoms. When multiple electrophilic carbons are present in a substrate, the most suitable electrophilic carbon atom will react the fastest and produce the major substitution product of the reaction.

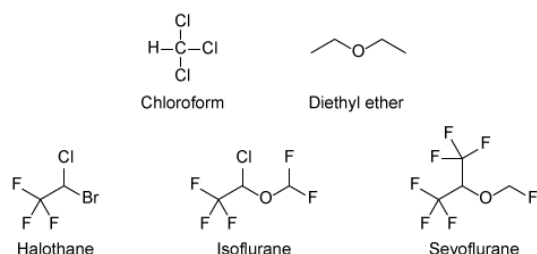
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Subject:
Organic Chemistry

QID:403863

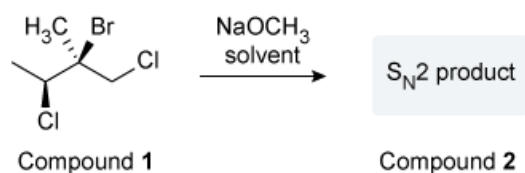
System:
Introduction to Organic Chemistry

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Scheme I

If Scheme I is repeated using $\text{NaOC}(\text{CH}_3)_3$ instead of NaOCH_3 , which of the following statements most accurately describes the outcome of the altered reaction?

- ☐ A. The reaction will occur slower because the electrophile is more sterically hindered. [9%]
- ☐ B. The reaction will occur faster because the electrophile is less sterically hindered. [1%]
- ☐ C. The reaction will occur faster because the nucleophile is less sterically hindered. [2%]
- ☒ D. The reaction will occur slower because the nucleophile is more sterically hindered. [1%]

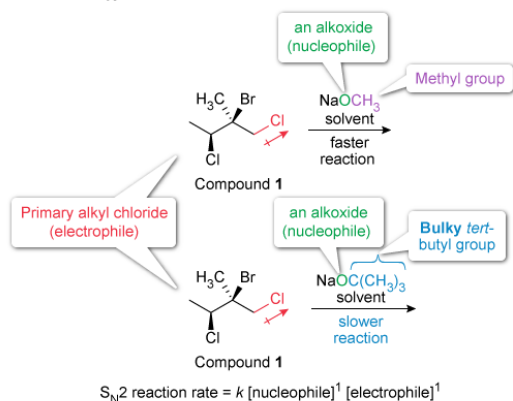
[87%]

Correct

86% answered correctly

Explanation:

S_N2 reactions with bulky nucleophiles



S_N2 reactions with a bulky nucleophile will occur slower due to decreased capacity for its lone pair of electrons to approach the electrophile

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An S_N2 reaction is a concerted bimolecular reaction where a nucleophile forms a new sigma bond to an electrophilic carbon and displaces a leaving group on the same carbon atom. Given that both events take place simultaneously, the nucleophile must approach the electrophile from the side opposite to the leaving group. This process is called a backside attack and leads to a **Walden inversion**.

As a bimolecular process, the steric effects of both the nucleophile and the electrophile will impact the rate of an S_N2 reaction. Sterically hindered molecules are structurally congested at or near the site of reaction and generally contain a greater number of atoms nearby that influence the rate of reaction. Given that an S_N2 reaction requires a bimolecular collision, the rate of reaction is faster for less-substituted electrophiles; tertiary electrophiles are unable to perform S_N2 reactions. Additional steric bulk limits the ability of the nucleophilic atom to approach the electrophile and donate its lone pair of electrons. **Sterically hindered nucleophiles** either have slower rates of S_N2 reaction or react differently (usually as a base) with the electrophile.

In this question, the sodium methoxide (1° alkoxide) is replaced by sodium *tert*-butoxide (3° alkoxide). As an electron-rich oxygen anion (alkoxide), these molecules are serving the role of **nucleophile** for the S_N2 reaction. The addition of three $-CH_3$ groups in proximity to the alkoxide oxygen atom **increases the steric bulk** of the nucleophile and limits its capacity to approach Compound 1 for a nucleophilic attack. As a result, the rate of the S_N2 will be **slower** with the new nucleophile.

(Choices A and B) The changes to the experimental procedure impact the structure of the alkoxide *nucleophile*. The structure of the *electrophile* (Compound 1) has not been altered.

(Choice C) The replacement of sodium methoxide with sodium *tert*-butoxide results in *additional* steric bulk that will limit its nucleophilic capacity. As a result, the rate of the S_N2

reaction is expected to *decrease* (not increase).

Educational objective:

The structure of a nucleophile can impact its steric hinderance and capacity to react with electrophiles. Bulky nucleophiles are less able to approach an electrophilic carbon atom during a substitution reaction and will lead to a slower rate of reaction.

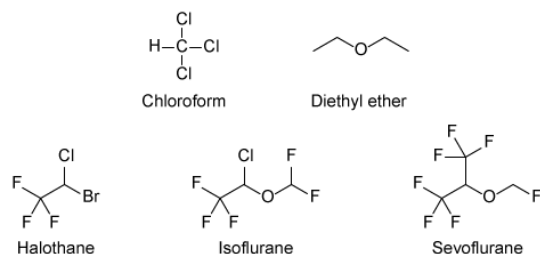
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Subject:
Organic Chemistry

QID:403864

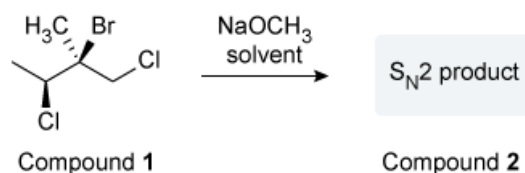
System:
Introduction to Organic Chemistry

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Scheme I

Based on the passage, which solvent will most likely promote the desired reaction outcome?

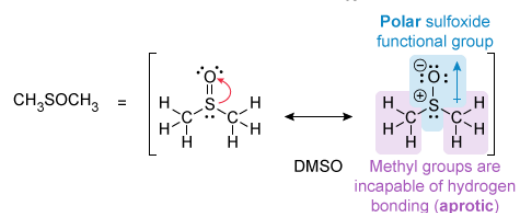
- ☐ A. $(\text{CH}_3\text{CH}_2)_2\text{O}$ [23%]
- ☐ B. $\text{CH}_3(\text{CH}_2)_4\text{CH}_3$ [19%]
- ☒ C. CH_3SOCH_3 [32%]
- ☐ D. $\text{CH}_3\text{CO}_2\text{H}$ [24%]

Correct

33% answered correctly

Explanation:

Polar aprotic solvents and S_N2 reactions



Polar aprotic solvents contain polar functional groups that lack the ability to hydrogen bond and are often used to promote S_N2 reactions.

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The solvent that is used for a nucleophilic substitution reaction can have a profound impact on the reaction outcome. Since first-order (S_N1) substitution reactions proceed through a carbocation intermediate, polar solvents are required to promote the formation and solvate the essential ionic intermediates. Many of the best solvents for S_N1 reactions are also **protic** and contain functional groups that are hydrogen bond donors. **Polar protic solvents** (eg, water, alcohols) tend to have higher relative boiling points to accommodate heated S_N1 reactions and, in many cases, double as the **nucleophile** for the substitution reaction.

Since second-order (S_N2) substitution reactions proceed through a polar transition state, polar solvents are also required. However, because S_N2 reactions require the use of strong nucleophiles, solvents that lack acidic hydrogen atoms (ie, **aprotic solvents**) are commonly used to ensure that the nucleophile is not surrounded by a rigid solvent shell. Common **polar aprotic solvents** include acetone (CH₃COCH₃), acetonitrile (CH₃CN), dimethylformamide (DMF, HCON(CH₃)₂), and dimethylsulfoxide (DMSO, CH₃SOCH₃).

Because the passage states that the transformation is intended to be an S_N2 reaction, the correct solution should be a polar aprotic solvent. **CH₃SOCH₃** contains a polar sulfoxide functional group (S=O) and lacks the ability to hydrogen bond, making it the best solvent of the given choices to promote an S_N2 reaction.

(Choice A) The molecule (CH₃CH₂)₂O (diethyl ether) contains an **ether** functional group. Ethers are relatively *nonpolar* and are not the best candidates to promote an S_N2 reaction.

(Choice B) The molecule CH₃(CH₂)₄CH₃ (hexane) is a nonpolar **alkane** that will be unable to solvate the polar reactants and transition state in an S_N2 reaction.

(Choice D) The molecule CH₃CO₂H (acetic acid) contains a **carboxylic acid** functional group.

Although carboxylic acids are polar, the acidic proton will rapidly neutralize the basic alkoxide nucleophile, greatly decreasing the efficiency of the S_N2 displacement.

Educational objective:

S_N2 reactions are promoted by polar aprotic solvents, which facilitate formation of the transition state without diminishing the strength of the nucleophile. Dimethylsulfoxide (DMSO) is a polar aprotic solvent.

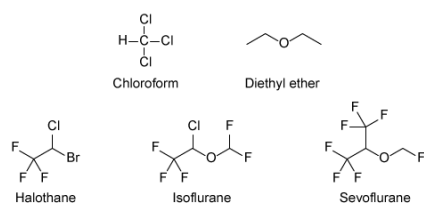
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Subject:
Organic Chemistry

QID:403865

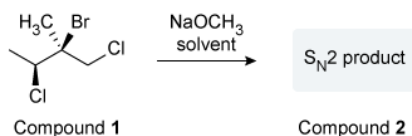
System:
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Scheme I

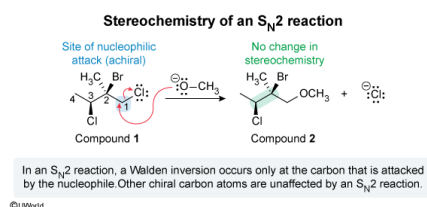
Which of the following structures depicts the product of the S_N2 reaction in the passage?

- ☐ A. [3%]
- ☒ B. [58%]
- ☐ C. [25%]
- ☐ D. [12%]

Correct

59% answered correctly

Explanation:



In a **substitution reaction**, a functional group on an atom is displaced by a different functional group. The lost functional group is the **leaving group**, named for its ability to leave with the pair of electrons from the sigma bond that connected it to the rest of the molecule. The leaving group polarizes this sigma bond to generate an electron-deficient carbon atom (an **electrophile**).

An **S_N2 reaction** is a second-order (bimolecular) nucleophilic substitution where the **nucleophile** displaces the leaving group in a single, concerted mechanistic step. The nucleophile approaches the electrophilic carbon from the side of the molecule opposite from the leaving group. If the electrophilic carbon atom is a chiral center, an S_N2 reaction will result in an inversion of stereochemistry at the electrophilic carbon atom, a process called the **Walden inversion**.

The passage describes an S_N2 reaction between Compound 1 (electrophile) and sodium methoxide (nucleophile). The primary alkyl chloride (carbon 1) has the least **steric hindrance** and will react the fastest. Since carbon 1 contains two identical hydrogen atoms, it is not a **chiral center**. Consequently, the S_N2 -facilitated Walden inversion will *not* result in an inversion of stereochemistry at carbon 1 of the product. The absolute configuration of all other chiral carbons in Compound 1 will remain *unchanged* in the product Compound 2.

(Choice A) This structure incorrectly shows inversion of stereochemistry at carbon 3 relative to Compound 1. Because carbon 3 does not undergo a Walden inversion in this reaction, its stereochemistry remains the same.

(Choice C) This structure depicts a product that has undergone a Walden inversion at all chiral carbons. An S_N2 reaction leads to a Walden inversion at *only* the carbon atom that is directly attacked by the nucleophile.

(Choice D) This structure incorrectly shows inversion of stereochemistry at carbon 2 relative to Compound 1. Because carbon 2 does not undergo a Walden inversion through this reaction, its stereochemistry remains the same.

Educational objective:

In an S_N2 reaction, nucleophilic displacement occurs with a Walden inversion of configuration at the attacked (electrophilic) carbon atom if the attacked carbon is chiral. The configuration of chiral centers at other locations within the structures of reactants will be unchanged.

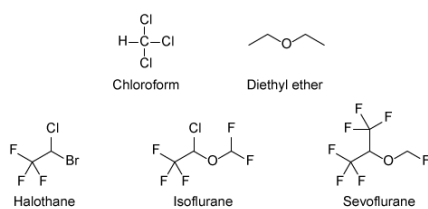
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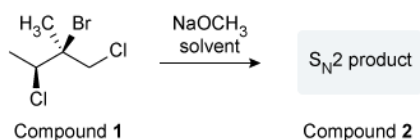
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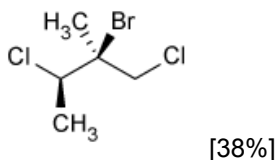
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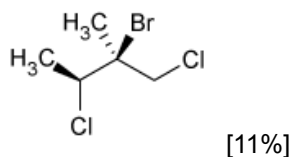
Scheme I

Which of the following structures does NOT depict the same molecule as Compound 1?

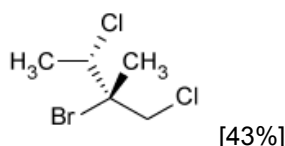
✓ ☒ A.



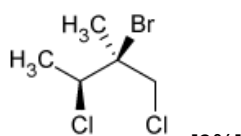
☐ B.



☐ C.

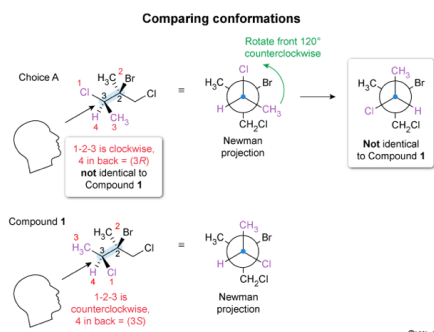


☐ D.



Correct

38% answered correctly

Explanation:

Conformations (conformational isomers) are different orientations of a single compound that differ due to the rotation of sigma bonds. The head-to-head overlap of orbitals in a sigma bond allows adjacent atoms to rotate *with respect to each other* to generate different conformations.

A **Newman projection** is a structural representation of molecular conformation. Groups on either the front or back atom can be rotated to generate a panel of **conformations**. Newman projections may also be used to determine whether two molecular depictions are conformations of the same molecule or different molecules.

This question asks which choice does *not* represent the same molecule as Compound 1. There are many ways to approach answering this question, including:

- Comparison of Newman projections
- Comparison of stereochemical configuration (*R/S* or *E/Z* designations)

In Choice A, the right portion of the structure is identical to the right portion of Compound 1. If the carbon 3-carbon 2 bond is rotated 120 degrees, the methyl group on carbon 3 matches the position and orientation of Compound 1. However, this rotation has the carbon 3 chlorine atom oriented into the plane (dash), instead of oriented out of the plane (wedge) as in Compound 1. Alternatively, the Newman projection of Choice A is *non-superimposable* with the Newman projection of Compound 1, and the **stereochemical assignment** of carbon 3 differs from the assignment of carbon 3 in Compound 1. Therefore, **Choice A** does *not* represent the same molecule as Compound 1.

(Choice B) **Counterclockwise rotation** of the sigma bond between carbons 3 and 2 generates a conformation that matches Compound 1. Alternatively, the (*S*) assignment for carbon 3 is identical to the assignment of carbon 3 in Compound 1.

(Choice C) The **Newman projection** for the sigma bond between carbons 3 and 2 can be rotated to generate an identical Newman projection as Compound 1. The configuration of carbon 3 is also (*S*), which matches the configuration of carbon 3 in Compound 1.

(Choice D) The carbon 1-carbon 2 sigma bond can be **rotated** 180 degrees to generate an identical structure as Compound 1.

Educational objective:

Conformations are different forms of the same molecule that differ due to the rotation of sigma bonds. Potential conformers can be compared using one of several methods, including Newman projections and stereochemical configuration.

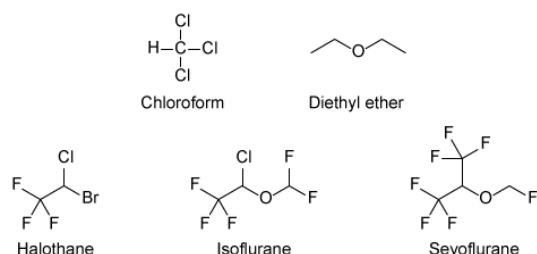
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Subject:
Organic Chemistry

QID: 403867

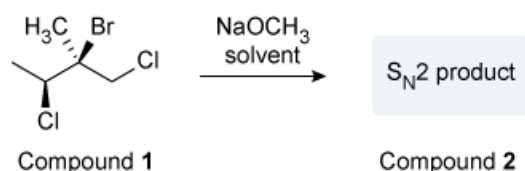
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Scheme I

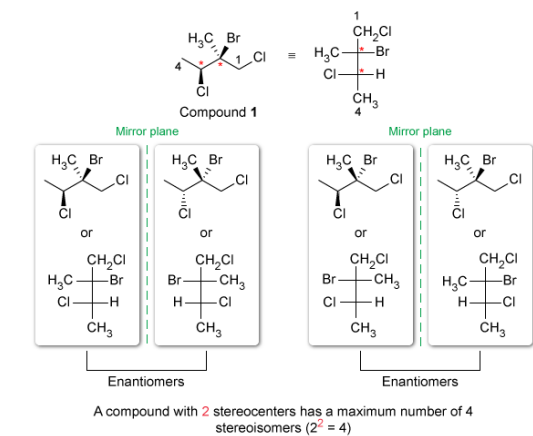
What is the maximum number of stereoisomers of **Compound 1** that exist?

- ☐ A. Two [10%]
- ☐ B. Three [4%]
- ☒ C. Four [78%]
- ☐ D. Eight [6%]

Correct
78% answered correctly

Explanation:

Calculating the maximum number of stereoisomers



Stereoisomers are isomers with the same connectivity of atoms but different orientation in three-dimensional space. The number of stereoisomers for a molecule is related to the number of **stereocenters** in the compound. The maximum number of stereoisomers possible for a compound is determined by the expression 2^n , where n is the **number of stereocenters**. Compounds that are nonsuperimposable mirror images are called **enantiomers**; any chiral material can have a maximum of only one enantiomer.

A **Fischer projection** is a two-dimensional method of drawing molecules containing sp^3 stereocenters. Each intersection of a Fischer projection represents one sp^3 stereocenter, where its horizontal lines are oriented toward the viewer, and its vertical lines oriented either in the plane or away from the viewer.

The number of stereoisomers may be lower than the maximum value if a compound contains an internal mirror plane of symmetry (making it a **meso compound**). Meso compounds are unique in that their mirror image *is* superimposable on itself. In this sense, meso compounds do not have an enantiomer and a structure with a meso compound has a lower total number of stereoisomers.

Carbon 2 and carbon 3 of Compound 1 both have four different substituents and are stereocenters. Since there are a total of two stereocenters ($n = 2$), Compound 1 has a **maximum of four stereoisomers** ($2^2 = 4$).

(Choice A) A compound with a maximum of two stereoisomers must have only one stereocenter ($2^1 = 2$ stereoisomers). Compound 1 has *two* stereocenters, one at carbon 2 and one at carbon 3.

(Choice B) It is not possible to have an *odd number* for the maximum number of stereoisomers, as this value is calculated by the expression 2^n .

(Choice D) In order to have a maximum of eight stereoisomers, a compound must contain *three* stereocenters ($2^3 = 8$). In Compound 1, since carbons 1 and 4 have at least two identical hydrogens attached, these positions are not stereocenters.

Educational objective:

Stereoisomers are isomers which differ in the way their atoms are held rigidly in three-dimensional space. The maximum number of stereoisomers for a structure can be calculated using the expression 2^n , where n is the number of stereocenters in the molecule.

Time spent:0 sec

Subject:
Organic Chemistry

QID:403868

System:
Introduction to Organic Chemistry